Week 2 – Workshop
Physical Properties, Mixing Laws, and Asking the Right Questions

Geophysics 3A: Potential Fields & Geothermics

July 7, 2026

Duration: 3 hours **Setting:** Workshop room

Preparation

Pre-reading: Course Notes Ch. 2

Lecture videos:

Notes:

Purpose

This workshop develops intuition for how physical properties (density ρ, thermal conductivity k, thermal diffusivity κ, bulk modulus K_T, shear modulus μ, seismic velocities V_P, V_S, magnetic susceptibility χ, and heat production A) depend on state variables (pressure P, temperature T, composition, porosity/cracks), and how geometry and fabric determine appropriate mixing relations (arithmetic, geometric, harmonic; Voigt–Reuss–Hill). Students practice cross-disciplinary collaboration and formulate the key questions a practicing geophysicist must ask a geologist before modeling or surveying.

Learning Outcomes

- Select and justify mixing laws based on rock fabric (random granular, layered/foliated, fractured) and target property (bulk vs. transport vs. elastic).
- Reason about how P–T–composition/porosity shift properties and create anisotropy.
- Identify which properties most strongly influence gravity, magnetic, and conductive heat-flow interpretations.
- Elicit geological information via well-posed questions to constrain non-uniqueness.

Reference concepts are aligned with the Week 2 notes on physical properties and mixing relationships.

Grouping

Create mixed groups (4–5 students each) with both geology and physics/engineering backgrounds.

Session Flow (170 min)

0–10 min: What does geophysics see?

Much like our senses, Different fields are sensitive to different things.

10–30 min: Mini-lecture and Q&A

- Bulk-additive vs. pathway-limited vs. elastic-interaction properties.
- Mixing relations: arithmetic (parallel), harmonic (series), geometric (random granular), Voigt–Reuss–Hill (elastic bounds).
- Scale-dependent anisotropy (micro/meso/macro fabrics).

30–45 min: Live Demonstration — "Crowd Conductivity"

Aim: Intuitively illustrate why different mixing laws arise and how anisotropy appears.

1. **Random scattering** (students distributed irregularly): volunteer transits across room. Multiple semi-obstructed pathways ⇒ geometric-like effective behavior.
2. **Parallel corridors** (lanes aligned with motion): fastest transit ⇒ arithmetic mean (parallel pathways).
3. **Series barriers** (rows perpendicular to motion): slowest transit ⇒ harmonic mean (series resistances).
4. **Anisotropy**: compare transit parallel vs. perpendicular to fabric.

Optional: time a handful of transits and note distribution shape differences.

45–65 min: Simulation Walkthrough

Builds on crowd conductivity with computation. We'll examine a pre-built Python visualization of transit times through three media (random, parallel, series) and the resulting histogram shapes. Relate to effective properties and mixing choices.

65–95 min: Activity 2.A: Photo-Based Anisotropy
(*micro–meso–macro classification*)

Distribute a set of photos (thin section, hand sample/outcrop close-up, outcrop/aerial). For each image, groups identify: geometry (random granular/layered/foliated/fractured), appropriate mixing law for density, conductivity, and moduli, whether the material is isotropic at that scale, and how conclusions change with scale.

95–155 min: Activity 2.B: Scenario Rotations (rich, real-world cases)

Each group tackles two scenarios (2̃0 min each: 12 min discussion, 5 min notes, 3 min question drafting). Use the dedicated scenario handouts. Outcomes must include: (i) property/mixing-law decisions and (ii) 2–3 *questions to ask the geologist* before committing to a model/survey.

155–170 min: Scenario Debrief

170–180 min: Q & A

Discussion of any new or unresolved questions.

180–190 min: Preview: Inversion

In geophysics, we typically measure fields and infer the underlying physical properties. In this course, however, we mostly focus on the *forward problem*: computing fields from a known distribution of properties.

In practice, we usually face the opposite challenge: the *inverse problem*. Inverse problems are difficult for several reasons:

- they are often *underdetermined*, with fewer observations than unknowns;
- *non-unique*, meaning different models can produce the same data; and
- *unstable*, where small changes in the model can produce large changes in the predicted field.

In addition, many geophysical relationships are non-linear, which adds further complexity.

Next week, we'll begin looking at how inversion methods address these challenges.

Workshop Notes

Workshop Notes

Activity 2.A
Physical Properties & Mixing Laws

Learning Objectives:

Learn how:

- geometry of materials affects bulk physical properties,
- how geometry necessitates the use of different mixing models,
- how scale affects what we perceive, and
- how to visually estimate the behavior of rock properties.

Background

Mixing relationships are mathematical methods that we use to approximate the physical behavior of an rock from its constituent mineralogy and porosity. These relationships can sometimes be derived from first principals—as we'll show for harmonic means and thermal conductivity later in the course—though more often they are empirical approximations that capture the messiness of the world.

Physical properties can be divided into two broad classes.

Bulk properties depend primarily on the amount of each constituent material present and are relatively insensitive to how those materials are arranged. Examples include density (ρ) and specific heat capacity (C_P). These properties are generally estimated using volume-weighted arithmetic averages.

Geometry-controlled properties depend not only on the constituent materials but also on how they are distributed and connected. Examples include thermal conductivity (k), electrical conductivity (σ), and many elastic properties (bulk and shear moduli). For these properties, the appropriate mixing relationship depends on the geometry and orientation of the materials.

Table 1: Guidelines for selecting mixing relationships.

Geometry / Fabric	Density	Conductivity	Elastic Moduli
Random granular mixture	Arithmetic	Geometric	Voigt–Reuss–Hill
Layered, parallel to transport	Arithmetic	Arithmetic	Direction-dependent
Layered, perpendicular to transport	Arithmetic	Harmonic	Direction-dependent
Strongly fractured or porous material	Arithmetic	Depends on connectivity; often harmonic-like	Depends on fracture geometry and filling

Instructions:

Open the slide deck (`isotropy_anisotropy.ppt`) on MyLearning course website. The pictures of rocks range from micrographs to aerial photos. For each photo, ask:

> If this were a gravity survey, would the distinction between individual minerals matter?
>
> If this were a conductivity problem, would connectivity matter?
>
> Consider how your interpretation might **change at a different scale**.

For each photo fill in your responses that:

- Identify the **fabric/geometry**.
- Identify the **approximate scale** of the photo.
- Select an appropriate **mixing law for density** (ρ).
- Select an appropriate **mixing law for thermal conductivity** (k).
- State whether the material is **isotropic or anisotropic** at this scale.

		Scale	**Mixing Law** (**A**)rithmetic, (**G**)eometric, (**H**)armonic		
Photo	**Description**	**Micro / Macro / Meso**	**Density**	**Thermal Conductivity**	**(I)sotropic / (A)nisotropic**
Example	uniform porous sandstone	Meso	A	G	I
#1					
#2					
#3					
#4					
#5					
#6					
#7					
#8					
#9					
#10					
#11					
#12					
#13					
#14					
#15					

Key Ideas

For bulk rock properties, an arithmetic mean is always used to compute the effective property estimate.

For pathway sensitive properties,

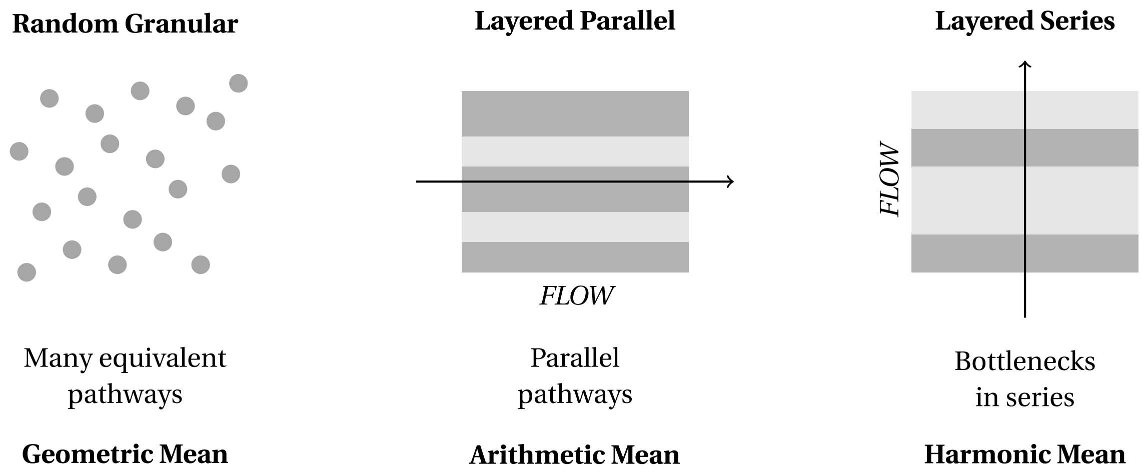

Figure 1: Conceptual relationship between fabric, transport geometry, and appropriate mixing relationship.

Activity 2.B

Modeling Property Scenarios: Choosing Physical Property Estimation Methods

Intructions

Below you will be provided with a set of scenarios that you as a geophysicist are required to develop a physical property model for. The point of this exercise is to develop a model that can be used as a reference based on the available information. In some cases the best and only option may be to run a geophysical survey and model the property from the results, though generally we can use existing samples and datasets to produce an initial property model.

For the discussions, students should be in groups mixed with geologists and physicists. For each scenario:

- draft 2–3 questions that a practicing geophysicist should ask the geologist for additional information about the geologic setting before devising your scenario,
- state your *preferred method* to obtain the required property (e.g., direct measurement, drill-core lab data, composition-based estimate, proxy/correlation, thermodynamic modeling, geophysical inversion/velocity–density regressions), and
- justify why you chose it over other options, list pros and cons, and write 2–3 key *questions to ask the geologist*.

Tip: articulate data availability, scale, P–T effects, alteration/weathering, and non-uniqueness when comparing methods.

Scenario A: Basin Density Model (Resources/Tectonics)

Motivation/Goal. You are supporting a basin analysis that informs *water resource assessment* and subsidence related to extraction. Drill cores are available at multiple wells, and a seismic stratigraphic framework defines a layered structure across the basin. The task is to build a depth-dependent **density model** suitable for forward gravity modeling and basin compaction studies. How will you use the available samples and geophysical data to develop the density model?

Your plan

2–3 Questions for the Geologist

Preferred method and why

Pros

- ______________________________
- ______________________________

Cons

- ______________________________
- ______________________________

Scenario B: Mountain-Belt Cross-Section (Tectonics)

Motivation/Goal. You are preparing a balanced structural cross-section across a metamorphic belt to support a tectonic reconstruction a for a large civil engineering project (construction of a hydroelectic dam). Rock types are mapped at the surface; access to samples are sparse and uneven, though the region has been explored for mineral resources previously. You need **density** for major units to test isostatic plausibility and gravity fit. How do you produce a density model?

Your plan

2–3 Questions for the Geologist

Preferred method and why

Pros

- ______________________
- ______________________

Cons

- ______________________
- ______________________

Scenario C: Upper-Mantle Density for Isostasy (Geodynamics)

Motivation/Goal. You are part of a team trying to better understand subduction dynamics. A 3-D regional tomography experiment images V_s and V_p within the mantle down to a depth of 410 km, capturing the mantle wedge, subducted plate and remaining upper mantle. The geodynamic model requires an upper-mantle **density** model as an input. What would be your approach to producing this density model?

Your plan

2–3 Questions for the Geologist

Preferred method and why

Pros

- ____________________
- ____________________

Cons

- ____________________
- ____________________

Scenario D: Aeromagnetic Susceptibility Model (Mineral/Exploration)

Motivation/Goal. For a greenfields **mineral exploration** program, you have a regional aeromagnetic survey. Your brief is to produce a **susceptibility model** that distinguishes key prospective units and to flag where remanent magnetization may invalidate induced-only assumptions.

Your plan

2–3 Questions for the Geologist

Preferred method and why

Pros

- ______________________________
- ______________________________

Cons

- ______________________________
- ______________________________

Scenario E: Heat-Flow Study in Granitic Terrane (Geothermal/Environmental)

Motivation/Goal. An energy agency is scoping geothermal potential and subsurface thermal risk for infrastructure. You must estimate **thermal conductivity,** $k(T)$ and **heat production** for granitic and metasedimentary units to forecast geotherms and transient responses. A number of pilot drill holes exist in the basin above, but do not interesct the granite. The only information about the granite comes from a gravity survey. How do you estimate the physical properties within the basin and granitic basement?

Your plan

2–3 Questions for the Geologist

Preferred method and why

Pros	**Cons**
• ____________________	• ____________________
• ____________________	• ____________________

Scenario F: Geothermal Heat Flow at the Base of the Antarctic Ice Sheet (Geothermal/Glaciology)

Motivation/Goal. You are tasked with developing a geothermal heat flow model into the base of an ice sheet to support assessment of the ice sheet stability due to recent climate change. Several coastal nunataks provide limited outcrops with physical specimens and geochemistry, but their small areal extent and clustered distribution offer only modest constraints on subsurface lithology and properties beneath the ice interior. There is a relatively low-resolution seismic volume for the region and high-resolution gravity data. How would you develop the necessary **thermal conductivity** and **heat production** models?

Your plan

2–3 Questions for the Geologist

Preferred method and why

Pros

- ____________________
- ____________________

Cons

- ____________________
- ____________________